



Project Title

LY Innovation-Based Major Project-B Report

Submitted in partial fulfillment of the requirements of the Degree of

Bachelor of Technology in Computer Engineering

by

Student Name1

Student Name2

Student Name3

Supervisor

Guide Name



Department of Computer Engineering

K. J. Somaiya Institute of Technology

An Autonomous Institute permanently affiliated to University of Mumbai

Ayurvihar, Sion, Mumbai -400022

2026-27



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CERTIFICATE



This is to certify that the project entitled “<Project Title>” is bonafide work of <Student Name 1, Student Name 2, Student Name 3> submitted to the University of Mumbai in partial fulfillment of the requirement in Major Project, for the award of the degree of “Bachelors of Technology” in “Computer Engineering”.

Guide Name
Project Guide
Department of Computer Engineering

Dr. Sarita Ambadekar,
Head, Department of Computer Engineering

Dr. Vivek Sunnapwar
Principal KJSIT

Place: Sion, Mumbai-400022
Date:

PROJECT APPROVAL FOR L. Y.

This project report entitled “<Project Title>” by

<Student Name1 (Roll No.)>

<Student Name2 (Roll No.)>

<Student Name3 (Roll No.)>

is an approved Last Year [Innovation-Based Major Project](#) in **Computer Engineering**.

Examiners:

1. _____

Name and Signature

External Examiner

2. _____

Name and Signature

Internal Examiner

Place: Sion, Mumbai-400022

Date:

DECLARATION

We declare that this written submission represents our ideas in our own words and where other's ideas or words have been included, we have adequately cited and referenced the sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Student Name1

Student Name2

Student Name3

Date:

ACKNOWLEDGEMENT

Before presenting our LY Major Project work entitled “<**Project Title**>”, we would like to convey our sincere thanks to the people who guided us throughout the course for this project work.

First, We would like to express our immense gratitude towards our **Project Guide <Name>** for the constant encouragement, support, guidance, and mentoring at the ongoing stages of the project and report.

We would like to express our sincere thanks to our **H.O.D. Dr. Sarita Ambadekar**, for the encouragement, co-operation, and suggestions progressing stages of the report.

We would like to express our sincere thanks to our beloved **Principal Dr. Vivek Sunnapwar** for providing various facilities to carry out this project.

Finally, we would like to thank all the teaching and non-teaching staff of the college, and our friends, for their moral support rendered during the course of the reported work, and for their direct and indirect involvement in the completion of our report work, which made our endeavor fruitful.

Place: Sion, Mumbai-400022
Date:

ABSTRACT

(Note: This is a Sample Abstract. Do not Copy)

In today's world, as businesses digitize and rely on cloud computing, the risk of cyberattacks rises. Malware infections, denial-of-service, and phishing are among the many types of attacks. Therefore, safeguarding data integrity and confidentiality and preventing harm to computer systems and networks through security measures is critical. An Intrusion Detection System (IDS) monitors network traffic to detect unauthorized access attempts. It can use two methods: signature-based, which identifies known threats based on pre-determined patterns, and anomaly-based, which analyzes traffic for unusual behavior. Anomaly-based detection can detect emerging threats, but it may also generate false positives and false negatives. To overcome the limitations of each method, a hybrid IDS approach has been proposed. A study introduced a hybrid IDS utilizing autoencoder-based anomaly detection and signature-based detection based on Random Forest and Light GBM classification models. The CICID2017 dataset, a standard IDS research benchmark dataset, was preprocessed, and autoencoders reduced its dimensions. The authors then trained the classification models using Random Forest and Light GBM, two commonly used algorithms for classification tasks. In addition, it proposes a second solution that uses autoencoders to detect anomalies. In this solution, the autoencoder is trained only on regular packets to help detect anomalies by calculating the anomaly threshold. Finally, Hybrid-IDS design by combining Snort3, Redis, and Elastic Stack is introduced.

To sum up, IDS is vital in identifying and stopping cyberattacks in today's digital landscape. While signature-based and anomaly-based detection have pros and cons, a hybrid IDS approach can leverage their benefits and mitigate their limitations. The suggested design, which employs Autoencoders, Random Forest and Light GBM classification models, can enhance the precision and efficiency of IDS.

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	1.2 Problem Definition	
	1.3 Aim and Objectives	
	1.4 Organization of the Report	
2	REVIEW OF LITERATURE	
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3	REQUIREMENT SPECIFICATION	
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4	PROJECT ANALYSIS & DESIGN	
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5		METHODOLOGY	
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7		RESULT ANALYSIS	
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		REFERENCES	
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		PLAGIARISM REPORT (Only First Few pages)	

Note:

1. This is a general template. The Sub -Headings/ contents of the Main chapters

- may change as per the requirements of your project type
2. Complete project report should be in Time New Roman font
 3. Font size: Heading -16, Subheading -14, text contents-12
 4. Spacing: Line spacing 1.5
 5. Text alignment: Justified
 6. Margin: Left- 1.5, other-1
 7. Minimum number of pages-50 (from Introduction to Conclusion)
 8. Maximum number of pages-70 to 80 (including Certificates at end)

LIST OF FIGURES

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LIST OF ABBREVIATIONS

Sr. No	Abbreviation	Description
1	IDS	Intrusion Detection System
2		
3		
4		
5		
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7		